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**Effects of Combining Electroacupuncture with General Anesthesia Induced by Sevoflurane in Patients Undergoing Supratentorial Craniotomy and Improvements in Their Clinical Recovery Profile & Blood Enkephalin**

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**ABSTRACT:**

Drug-induced anesthesia combined with electroacupuncture (EA) in patients has been put into practice in recent years in China. In this study, we showed the effectiveness of EA on the speed of post-operative recovery of patients undergoing supratentorial craniotomy and the potential clinical mechanism of EA. Dual channel electrical stimulator made by HANS Beijing connected the following acupoints respectively: LI4 (Hegu), SJ5 (Waiguan), ST36 (Zusanli), BL63 (Jinmen), LR3 (Taichong), and GB40 (Qiuxu). Disperse-dense and

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**Keywords:** Glioma; Craniotomy; Electroacupuncture; Analgesia; Sevoflurane; Anesthesia; Recovery time after surgery; Enkephalin.

INTRODUCTION

Gliomas are the most serious type of brain tumors in adults, including astrocytic tumor、oligodengdroglial tumors、mixed gliomas and ependymal tumors. They originate in neuroectoderm, grow in an uninhibited fashion, diffusely infiltrate the brain, make surgical resectioning difficult or even impossible. Among them, glioblastoma multiforme (GBM) is the most common form of malignant brain cancer in adults. Patients with GBM have a uniformly poor prognosis, with a median survival of one year[1]. Surgical resection is the most effective therapeutic method to date. Despite the availability of newer analgesic drugs and techniques, concerns remain on the side effects of both opioid and nonopioid analgesic techniques, such as nausea and vomiting, hypotension, urinary retention and ileus, pruritis, constipation and even respiratory depression. For patients undergoing craniotomy, these side effects may lead to devastating results owing to the increased risk of intracranial hemorrhage and neurological dysfunction [2]. The incidence of side effects is related in part to the total dose of opioid analgesic medication [3]. These side effects can be reduced by using minimal flow rates of the opioid dosage but cannot be completely eliminated due to the need of analgesic medication [4]. Acupuncture is a complementary non-pharmacological therapy with few side effects of analgesic medication. Its importance for health care in Asian culture can be traced back almost 3000 yrs, and it has been practiced to treat a variety of pathological symptoms and conditions [5]. Acupuncture in conjunction with electrical stimulation techniques (e.g., electroacupuncture, transcutaneous electrical nerve stimulation [TENS]) are based on traditional Chinese analgesic methods. EA is an inexpensive, safe, and easy stimulation technique that has been used for the management of pain following a wide variety of surgical procedures [6,7]. The present study was randomize, sham controlled, and blinded design to objectively test the hypothesis that EA would: (1) decrease the amount of sevoflurane needed for anesthesia; (2) facilitate regulation of the physiological and pathological changes of the human body during surgery; and (3) improve the overall recovery profiles for the post-operative patient. Further, we have assayed serum levels of enkephalin to assess physiological changes in response to EA.

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MATERIALS AND METHODS

Eighty patients having an American Society of Anesthesiologists (ASA) classification of physical status I - II and with supratentorial tumors were recruited for this study. Written informed consents were obtained from all patients. The study protocol was approved by the Institutional Review Board at Beijing Tiantan Hospital. Equal numbers of patients were randomly assigned to the sham EA-group and the EA- group. The acupoints included for the study were LI4 (Hegu), SJ5 (Waiguan), ST36 (Zusanli), BL63 (Jinmen), LR3 (Taichong), GB40 (Qixu) (Fig. 1).

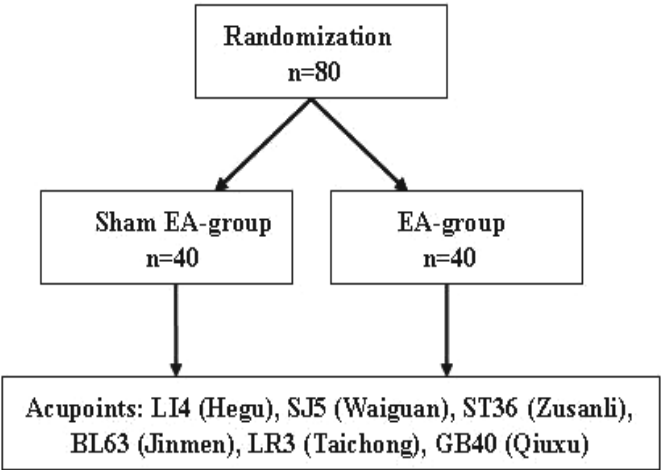
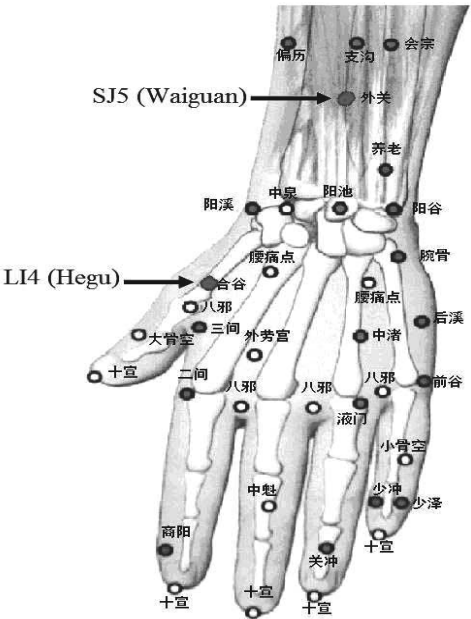


Fig. 1A: The groups and acupoints of the study.



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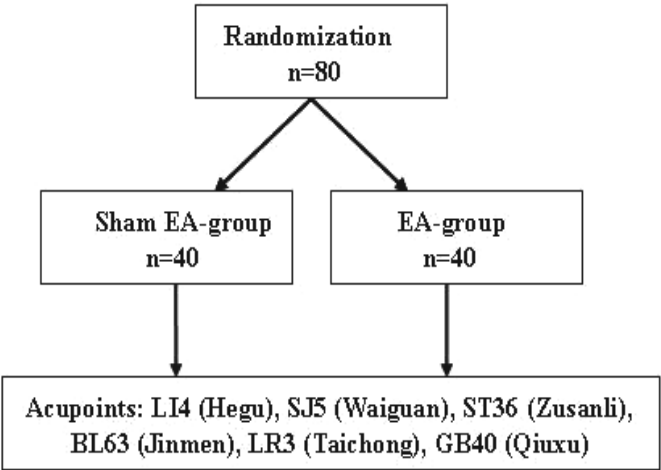


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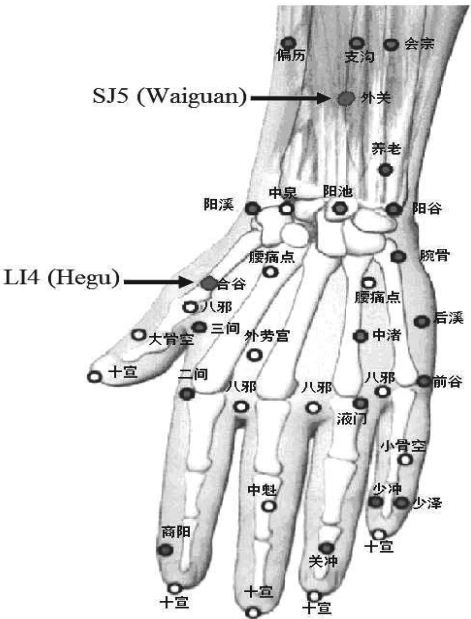


Fig. 1B: The locations of LI4 (Hegu) and SJ5 (Waiguan): Hegu (LI4), on the dorsum of the hand, between the first and the second metacarpal bones, on the radial side of the midpoint of the second metacarpal bone. Waiguan (SJ5), on the dorsum of the forearm, 2 cun proximal to the dorsal wrist crease, between the radius and the ulna.

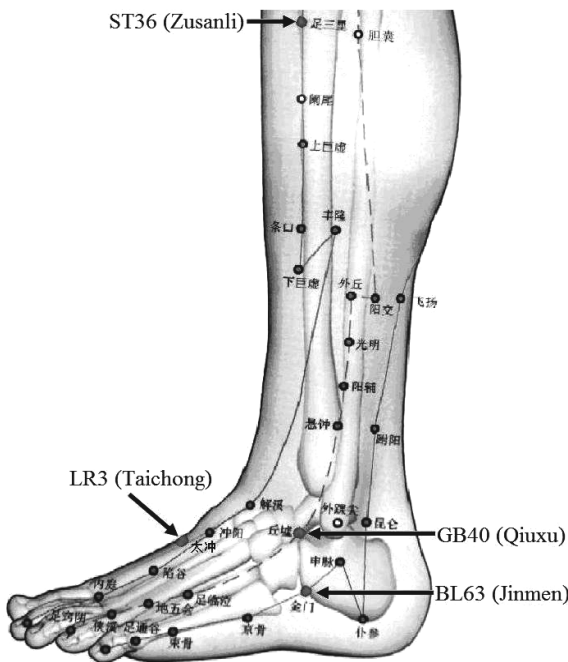


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Excluded from the study were: (1) pregnant and/or lactating women; (2) patients with clinically significant cardiovascular problems or pulmonary diseases; (3) long term smokers; (4) obese patients (BMI > 30); (5) emergency patients; and (6) patients with bleeding > 2,500 mL during surgery and/or patients having a surgical duration > 8 h.

Anesthesia was induced with 0.3  $\mu$ g/kg sufentanil i.v.and 2 mg/kg propofol i.v. Tracheal intubation was facilitated with 0.1 mg/kg vecuronium. Ventilation was controlled to maintain end-tidal PaCO<sub>2</sub> at 28-33 mm Hg after intubation. General anesthesia was initially maintained with sevoflurane 2% in oxygen 1L/Min. Vecuronium top-up doses were administered as guided by TOF monitor (TOF watch SX monitor). Before incision,0.5% ropivacaine was used for scalp infiltration, 0.15  $\mu$ g/kg sufentanil was administered before fixing the head brace and after

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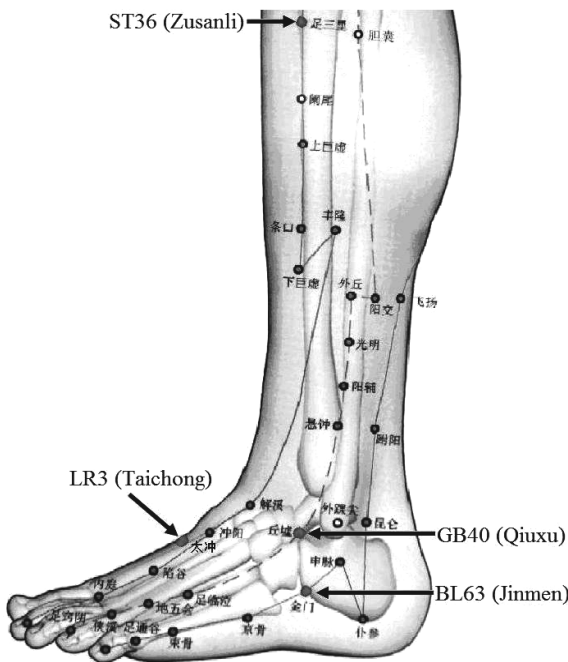


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A trained acupuncturist determined the locations for acupoints: LI4 (Hegu), SJ 5(Waiguan), ST36 (Zusanli), BL63 (Jinmen), LR3 (Taichong), GB40 (Qixu) ,based on the Traditional Chinese Medicine meridian theory. The needles were inserted at the specific points (17–33 mm depth, depending upon the build of patient) in the electroacupuncture group. A De Qi sensation was ensured by attaching small crocodile clips to the end of the needles to connect them to an electroacupuncture device (HANS dual-channel unit; LH202H, Huawei, Beijing) [8] (Fig. 2).

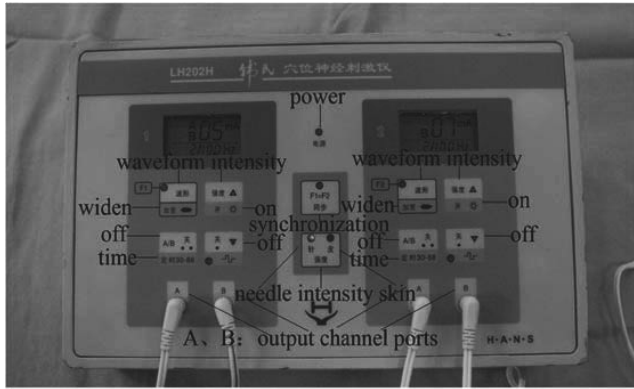


Fig. 2A: HANS dual-channel unit, LH202H, rechargeable batteries electrical stimulator made in Beijing, China.



Fig. 2B: Patient's acupoints: LI4 (Hegu) and SJ5 (Waiguan) connected to HANS electrical stimulator in EA-group. Small crocodile clips attached to the end of the needles to connect them to an electroacupuncture device.

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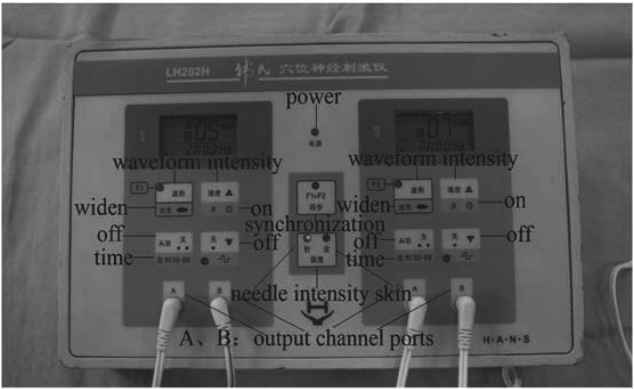


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The electroacupuncture device was a biphasic pulse generator and was used with a maximum tolerable intensity of current and frequency: disperse-dense waves, frequency of waves(pulse rates)were 2 Hz/ 100 Hz[9], altered / 3second, pulse duration was 0.6ms/0.2ms; 2 Hz : 0.6ms; 100 Hz: 0.2ms; symmetric biphasic pulse wave (Fig. 3).

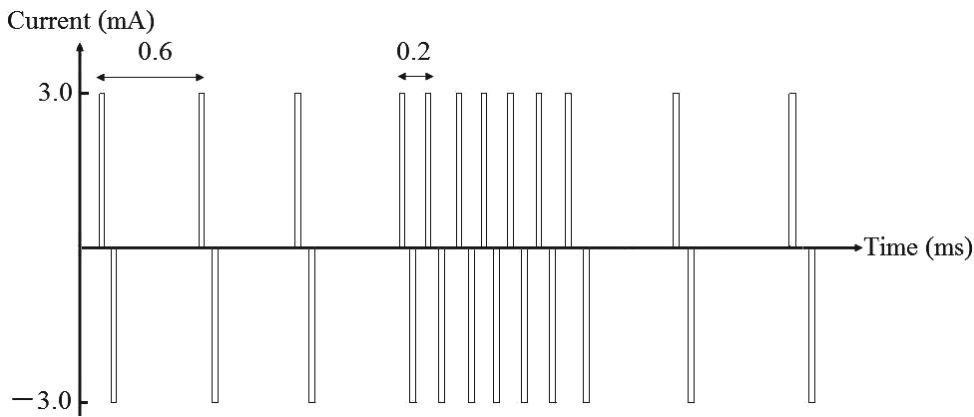


Fig.3: Diagram of the waveform of electroacupuncture.

The acupuncture was applied and activated before the induction of anesthesia in each EA patient, and was maintained until completion of the operation (completion of skin closure). In controls (the sham EA- group), the electric wire was placed at the surface of the chosen acupoint sites, however, no needles were inserted or current applied.

The end-tidal sevoflurane concentration and the speed of recovery were recorded, including the autonomous respiration recovery time (ARRT), tracheo-tube removal time (TTRT), eye opening time (EOT), voluntary motion recovery time (VMRT), orientation force recovery time (OFRT), and operating room departure time (ORDT). The start point for timing the above events was the cessation of sevoflurane administration for each patient in the 2 groups.



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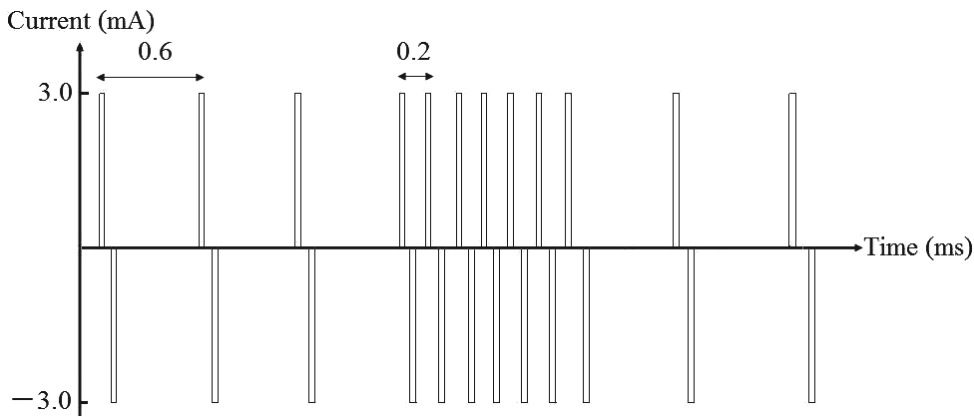


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A five milliliter blood sample was drawn from each patient using the femoral vein under aseptic conditions, during general anesthesia at 0, 30, 45, 60, 120 and 240 min. Blood samples were immediately transferred to vacutainers and centrifuged to facilitate serum collection. Serum was transferred to storage vials and kept at -80°C until analysis. Quantification of serum enkephalin was performed using ELISA, i.e. (Human Enkephalin Elisa Kit. Catalog No:E01E0312). Valid samples were 13 and 17 in control and EA-group respectively.

The data were entered into EXCEL data sheets and analyzed using the SPSS 13.0 statistical package. Data from the control and EA groups (sevoflurane requirement and the recover profiles) were compared using the Student *t* test. The time-dependent intragroup data for serum enkephalin were evaluated by repeated-measures and analysis of variance. *P* < 0.05 was considered statistically significant. Data were expressed as the mean ± SD.

RESULTS

In this study no patient had previously received an EA treatment. All patients were enrolled based on the inclusion criteria (above) and were randomly assigned to the two groups. No complications related to EA procedures were observed during the operation. The two treatment groups were comparable with respect to both demographic and clinical variables. There were no significant intergroup differences in patients' age, weight, height, or anesthesia time (Table 1).

| General conditions                                        | Sham EA-group (n=40) | EA- group (n=40) |
|-----------------------------------------------------------|----------------------|------------------|
| Age (years)                                               | 39±11                | 43±11            |
| Weigh (kg)                                                | 66±10                | 69±15            |
| Male/female (cases)                                       | 21/ 19               | 17/ 23           |
| Anesthesia time (min)                                     | 211±56               | 233±56           |
| Meningioma                                                | 17                   | 11               |
| Glioma                                                    | 11                   | 23               |
| Metastatic tumor                                          | 2                    | 6                |
| Intraoperative sufentanil (µg)                            | 38±4                 | 37±1             |
| Data presented as mean±SD or as a number (n) of patients. |                      |                  |

Table 1: Patients' characteristics and perioperative use of analgesics.

The consumption of sevoflurane during operation was shown in Table 2.

| Stages of surgery    | Sevoflurane end-tidal concentration in Sham EA-group (%); n=40 | Sevoflurane end-tidal concentration in EA-group (%); n=40 | Sham EA-group vs EA-group sevoflurane reduction (%) | P      |
|----------------------|----------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------|--------|
| Skin incising        | 1.60 ± 0.37                                                    | 1.50 ± 0.26                                               | 6.25                                                | 0.263  |
| Skull drilling       | 1.72 ± 0.43                                                    | 1.58 ± 0.32                                               | 8.13                                                | 0.175  |
| After skull drilling | 1.74 ± 0.39                                                    | 1.57 ± 0.31                                               | 9.19                                                | 0.032* |

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|                             |             |             |       |        |
|-----------------------------|-------------|-------------|-------|--------|
| After skull opening         | 1.75 ± 0.39 | 1.58 ± 0.30 | 9.19  | 0.035* |
| Dura incising               | 1.79 ± 0.43 | 1.58 ± 0.28 | 11.73 | 0.014* |
| Intracranial surgery 10 min | 1.78 ± 0.45 | 1.56 ± 0.26 | 12.35 | 0.010* |
| Intracranial surgery 20 min | 1.76 ± 0.44 | 1.54 ± 0.26 | 12.50 | 0.010* |
| Intracranial surgery 30 min | 1.80 ± 0.44 | 1.59 ± 0.28 | 11.66 | 0.022* |
| Dura suturing               | 1.83 ± 0.39 | 1.70 ± 0.28 | 7.10  | 0.110  |
| Skin suturing               | 1.73 ± 0.37 | 1.59 ± 0.23 | 8.09  | 0.620  |
| *P<0.05                     |             |             |       |        |

Table 2: The consumption of sevoflurane during operation. 9.62% decreased in sevoflurane requirement were produced in EA-group compared to sham EA-group. ARRT, TTRT, EOT, VMRT, OFRT and ORDT after stopping administration of sevoflurane in EA group were all less than in control group.

The recovery profiles were shown in Table3.

| Recovery of anesthesia | Sham EA-group | n  | EA-group | n  | Sham EA-group vs EA-group sevoflurane reduction (%) | P      |
|------------------------|---------------|----|----------|----|-----------------------------------------------------|--------|
| RRT                    | 12.69         | 32 | 8.14     | 29 | 35.86                                               | 0.013* |
| TTRT                   | 21.32         | 31 | 15.55    | 29 | 27.07                                               | 0.005* |
| EOT                    | 28.52         | 31 | 17.57    | 28 | 38.38                                               | 0.000* |
| VMRT                   | 29.47         | 30 | 20.59    | 27 | 30.11                                               | 0.002* |
| OFRT                   | 34.17         | 30 | 22.22    | 27 | 34.95                                               | 0.000* |
| ORDT                   | 39.32         | 31 | 28.00    | 28 | 28.80                                               | 0.000* |
| *p < 0.05              |               |    |          |    |                                                     |        |

Table 3: The recovery profiles (min): ARRT: autonomous respiration recovery time; TTRT: tracheo-tube removal time; EOT: eye-opening time; VMRT: voluntary motor recovery time; OFRT: orientation force recovery time; and ORDT: operating room departure time.

Blood enkephalin, a pain relieving neurotransmitter, was consistently higher in the control than in the EA-group during the operative period according to ELISA measurements (P=0.011; Fig. 4).

|                             |             |             |       |        |
|-----------------------------|-------------|-------------|-------|--------|
| After skull opening         | 1.75 ± 0.39 | 1.58 ± 0.30 | 9.19  | 0.035* |
| Dura incising               | 1.79 ± 0.43 | 1.58 ± 0.28 | 11.73 | 0.014* |
| Intracranial surgery 10 min | 1.78 ± 0.45 | 1.56 ± 0.26 | 12.35 | 0.010* |
| Intracranial surgery 20 min | 1.76 ± 0.44 | 1.54 ± 0.26 | 12.50 | 0.010* |
| Intracranial surgery 30 min | 1.80 ± 0.44 | 1.59 ± 0.28 | 11.66 | 0.022* |
| Dura suturing               | 1.83 ± 0.39 | 1.70 ± 0.28 | 7.10  | 0.110  |
| Skin suturing               | 1.73 ± 0.37 | 1.59 ± 0.23 | 8.09  | 0.620  |
| *P<0.05                     |             |             |       |        |

Table 2: The consumption of sevoflurane during operation. 9.62% decreased in sevoflurane requirement were produced in EA-group compared to sham EA-group. ARRT, TTRT, EOT, VMRT, OFRT and ORDT after stopping administration of sevoflurane in EA group were all less than in control group.

The recovery profiles were shown in Table3.

| Recovery of anesthesia | Sham EA-group | n  | EA-group | n  | Sham EA-group vs EA-group sevoflurane reduction (%) | P      |
|------------------------|---------------|----|----------|----|-----------------------------------------------------|--------|
| RRT                    | 12.69         | 32 | 8.14     | 29 | 35.86                                               | 0.013* |
| TTRT                   | 21.32         | 31 | 15.55    | 29 | 27.07                                               | 0.005* |
| EOT                    | 28.52         | 31 | 17.57    | 28 | 38.38                                               | 0.000* |
| VMRT                   | 29.47         | 30 | 20.59    | 27 | 30.11                                               | 0.002* |
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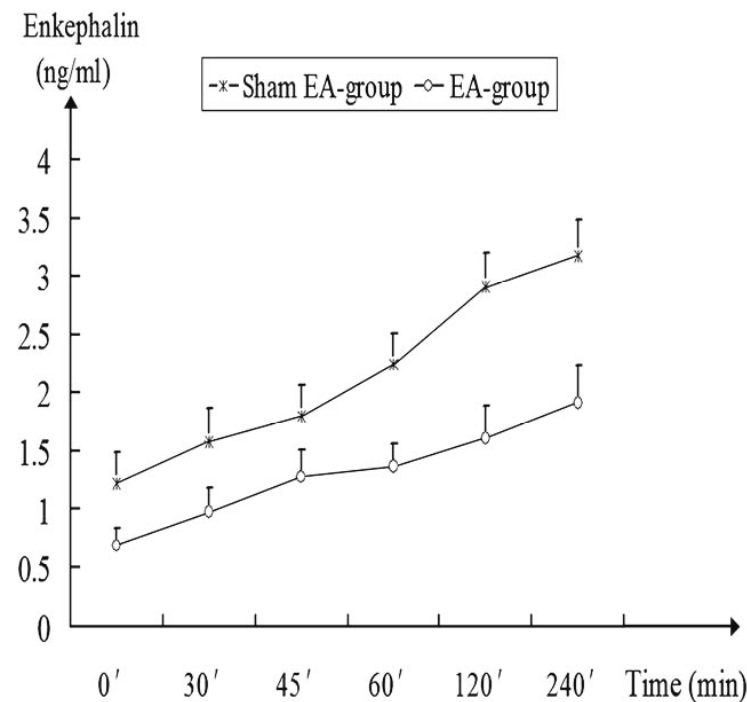


Fig. 4: Detected enkephalin in EA-group and sham EA-group. Enkephalin was consistently higher in sham EA-group than in EA-group during the operative period according to ELISA measurements ( $P=0.011$ ).

### DISCUSSION

Omura [10] reports that the best pulse rates for average human adult is same as pulse rate of the patient which is synchronized with local blood circulation between 1 and 1.5 pulses / sec. Any pulse rate over 2 pulses/sec in human adult creates various degrees of local circulatory disturbance and reduces pain relief time duration depending on pulse repetition rate and other stimulating parameters. For prolonged (post-stimulation) effects, typically equating to 1.2-3pps [11-13]. Richard P. Francis [14] believes that the characteristics of acupuncture-like TENS include pulse rate between 1-5pps; a pulse duration between 100 $\mu$ s and 200 $\mu$ s. During our study, according to the release of analgesic substances and synergistic effect, we selected 2Hz/100 Hz frequency to work. Han JS [9] indicated that enkephalin and endorphin could be released with 2Hz electric stimulation, and the release of dynorphin need 100Hz electric stimulation. 2Hz EA produced a 7 fold increase in met-enkephalin [15]. Radioimmunoassay has shown that rat brain levels of beta -endorphin are lower than those of the enkephalins [16]. Enkephalin within vagal nuclei of the medulla might play a role in visceral reflexes affected by opiates, such as the cough and vomiting reflexes. Within the periaqueductal gray, enkephalin may mediate electrical stimulation-induced analgesia [17-19]. Enkephalin in the infundibulum and periventricular nuclei of the hypothalamus may reflect loci where opiates influence endocrine functions. Within certain limbic and cortical areas, especially the central nucleus of the amygdala, enkephalin might be

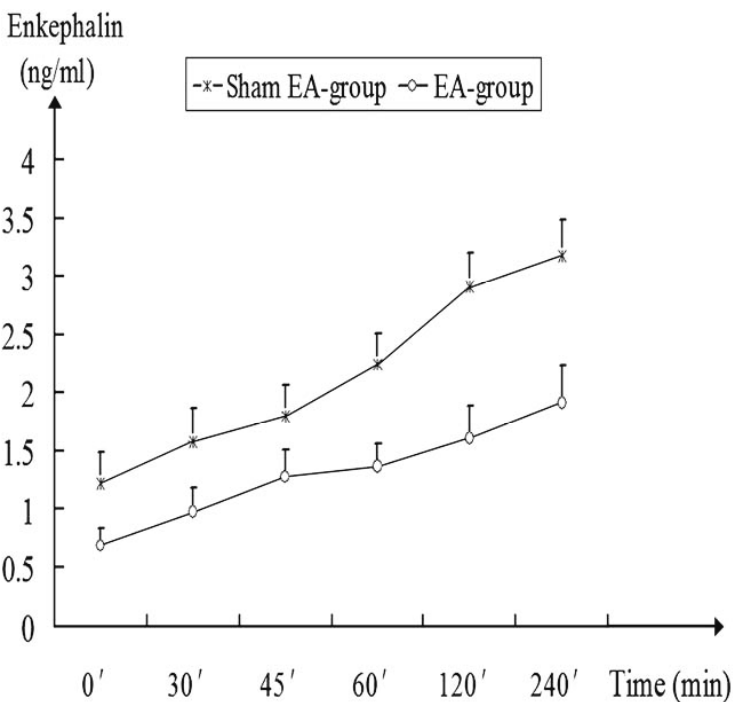


Fig. 4: Detected enkephalin in EA-group and sham EA-group. Enkephalin was consistently higher in sham EA-group than in EA-group during the operative period according to ELISA measurements ( $P=0.011$ ).

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associated with the euphoric effects of opiates[16]. Beta -endorphin is a potent analgetic, being 18 to 33 times a more potent than morphineon a molar basis. Chronic, localized infusion of beta-endorphin into the rat brain can induce morphine-like physical dependence [20]. Many brain nuclei composing a complicated network are involved in processing acupuncture analgesia, including the nucleus raphe magnus (NRM), periaqueductal grey (PAG), locus coeruleus, arcuate nucleus (Arc), preoptic area, nucleus submedius, habenular nucleus, accumbens nucleus, caudate nucleus, septal area, amygdale, etc[21]. The result of our study confirmed the release of serum enkephalin. More research will continue to study the effects of 100Hz electric stimulation on circulatory disturbance.

This study suggests that EA at the acupoints (Hegu, Waiguan, Zusanli, Jinmen, Taichong, and Qiu xu) combined with sevoflurane could improve recovery profiles of patients undergoing supratentorial craniotomy. Patients from EA group required 9.62% less sevoflurane than the control patients during surgery ( $p<0.05$ ). It indicates that EA had effect of anesthesia. Acupuncture analgesia is the basis of acupuncture anesthesia [22-26]. Previous studies have shown the effects of EA on painful, such as: neck pain, shoulder pain, elbow pain, low back pain, knee pain etc [27-30]. EA may be a complementary method for anesthesia that could minimize the side effects from anesthetics. However, EA cannot replace the role of anesthetics. Our results show that patients' recovery profiles of EA, including autonomous respiration recovery time (ARRT), tracheotube removing time (TTRT), eye opening time (EOT), voluntary motion recovery time (VMRT), orientation force recovery time (OFRT), and operating room departure time (ORDT) after stopping administration of sevoflurane, are all less than control group. These effects may be related to the decreased sevoflurane requirement to maintain anesthesia. The most important function about EA may be the regulation for body.

Acupuncture is an intricate therapeutic system. Its mechanisms include traditional Chinese theories along with western anatomy and physiology. Traditional Chinese acupuncture presumes that there are two opposing and complementary forces that coexist in nature: Yin and Yang. These two forces interact to regulate the flow of "vital energy" known as Qi. It is defined that good health is a balance between yin and yang, in this situation, Qi flows freely through a system of channels (meridians), which bring Qi from the internal organs to the skin's surface. Along these meridians there are acupuncture points that can be stimulated to correct the imbalance and restore the body to normal health. In disease this flow is disrupted causing an imbalance between yin and yang. Needles inserted into specific points along meridians are said to regulate imbalances in the flow of Qi and treat diseases. According to the experiences of curing headaches using acupuncture, the traditional Chinese theories propose that: distant acupoints treat proximal disease. We have chosen the distant acupoints: Hegu, Waiguan, Jinmen, Taichong, Zusanli and Qiu xu to insert needles. Acupoints usage is effective for treatment, the traditional Chinese explanation has little "scientific" basis. Western scientific explanations of EA are based on the presumption that acupuncture induces signals in afferent nerves that modulate spinal signal transmission and pain perception in brain. Neural mechanisms that have been proposed for short-term analgesic effects of acupuncture include spinal gating of descending inhibitory pathways. Spinal gating pain is mediated by two types of peripheral fibers: myelinated A-fibers responsible for sharp, well-localized pain ('fast'pain) and smaller unmyelinated C-fibers responsible for prolonged and ill-defined pain ('slow'pain) [31]. EA stimulates the large A, $\beta$ -fibers and activates the gating mechanism. This may inhibit central pain transmission

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through enkephalin dependent mechanisms [32,33]. Peripheral noxious stimulus, such as pinpricks, results in impulses ascending in the spinothalamic tracts to the thalamus. At the midbrain, collateral pathways project to the periaqueductal grey area (PAG). Electrical stimulation of the PAG leads to analgesia that is reversed by naloxone [34]. Neurons from the PAG project to the nucleus raphe magnus of the medulla and from there serotonergic fibers descend to the stalked cells of the dorsal horn where release of enkephalin inhibits SG (spiral ganglion) cells [32,35], thereby blocking the centripetal transmission of pain from C-fibers. Pomeranz proposed that acupuncture stimulation activates A $\delta$ -and C afferent fibers in muscle, causing signals to be transmitted to the spinal cord and results in release of dynorphin and enkephalins. Pomeranz's theory was confirmed by a large series of experiments in his research laboratory and other investigators [36-41]. The regulation of enkephalin peptide synthesis was mainly at the mRNA level [42] and EA was shown to enhance preproenkephalin mRNA expression in rostral ventrolateral medulla of rats [43]. It suggests that there would be more peptides to produce following EA [44]. In our study, enkephalin did not increase in EA-group compared with sham EA-group. The reasons maybe include the flowing items:(1) the blood samples detected enkephalin could not consistent with the quantity of EA-group (n=40) and sham EA-group(n=40) repectively; this maybe the limitation of our study; (2) the exact mechanisms of anesthetics remain unclear to date; maybe the change of enkephalin in EA-group is the result of the combination EA and anesthetics. Our results indicate that dynamic change of enkephalin during operations in EA is different from control. The tendency of increased enkephalin in EA suggests that enkephalin might play a role for general anesthesia in pain reduction.

CONCLUSION

EA anesthesia could not replace anesthetics completely. However, EA has become the most popular complementary and alternative medicine modality and is highly praised by physicians and anesthetists. Due to the limited number of blood samples available to us, future research maybe needed to assess enkephalin and mechanisms of EA treatment.

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